

Prepare: Textbook §3.4

Primary objective. Approximate the flux spectrum $\phi(E)$ of fast and epithermal neutrons subject to elastic collisions without absorption.

1 Review: why do we need a spectrum?

Last time, $\eta(E)$ was a first step toward k : an energy average would give a single value approximately bounding k from above. Before that, we developed the energy-transfer distribution $P(E \rightarrow E')$.

The missing weighting function

The distribution of neutron energies comes from competition between **scattering and absorption**. What weighting function should we use when averaging over energy?

2 Neutron density, flux, and reaction rates

All particle counts, distances, and interaction counts below are **expected** quantities.

Density and flux distributions

$n'''(E)$ is the neutron density per unit energy, with units $\text{cm}^{-3} \text{eV}^{-1}$. The **neutron flux** is

$$\phi(E) = n'''(E) v(E), \quad v(E) = \sqrt{\frac{2E}{m}}.$$

Here m is the neutron mass and $E = \frac{1}{2}mv^2$. The units of $\phi(E)$ are $\text{cm}^{-2} \text{s}^{-1} \text{eV}^{-1}$.

Path-length interpretation. $\phi(E) dE$ is the total distance traveled in one second by all neutrons with energies between E and $E + dE$ in a 1 cm^3 volume. This interpretation is especially useful for **Monte Carlo simulation**.

Differential and integrated reaction rates

$$\phi(E) \Sigma_x(E) dE$$

is the number of interactions of type x , per second per cm^3 , for neutrons in the same energy interval. Integrating over energy gives

$$R_x = \int_0^\infty \Sigma_x(E) \phi(E) dE, \quad [R_x] = \text{cm}^{-3} \text{s}^{-1}.$$

Units and meaning

What distinguishes $\phi(E)$ from $\phi(E) dE$? What does multiplying by $\Sigma_x(E)$ count?

3 Balance at a neutron energy E

The **total interaction rate density** is

$$\Sigma_t(E) \phi(E) \quad [\text{cm}^{-3} \text{s}^{-1} \text{eV}^{-1}].$$

Each interaction removes a neutron from energy E . To balance that loss, neutrons must scatter into E , be born at E by fission, or arrive from an external source.

The slowing-down (or spectrum) equation

$$\begin{aligned} \Sigma_t(E) \phi(E) = & \int_0^\infty P(E' \rightarrow E) \Sigma_s(E') \phi(E') dE' \\ & + \chi(E) \int_0^\infty \bar{v}(E') \Sigma_f(E') \phi(E') dE' \\ & + s_{\text{ext}}(E). \end{aligned}$$

Like FNRP 3.15

The right-hand terms describe **in-scattering**, **fission births**, and an **external source**, respectively. The energy-transfer cross section is

$$\Sigma_s(E' \rightarrow E) = P(E' \rightarrow E) \Sigma_s(E').$$

This balance follows from the energy-dependent diffusion equation when space and time are eliminated.

4 Three energy ranges

Approximations by energy range

Fast	0.1–10 MeV; $\chi(E) > 0$. $\Sigma_s(E' \rightarrow E) = 0$ for $E' < E$.
Epithermal	1 eV–0.1 MeV; $\chi(E) = 0$. $\Sigma_s(E' \rightarrow E) = 0$ for $E' < E$.
Thermal	10^{-3} –1 eV; $\chi(E) = 0$. $\Sigma_s(E' \rightarrow E) \neq 0$ can occur for $E' < E$.

Why can thermal neutrons gain energy? Their kinetic energies can be comparable to those of surrounding nuclei, so a collision can transfer energy to the neutron.

Why neglect low-energy fission births? Most fission neutrons have energies around 1–2 MeV; very few are born below 0.1 MeV.

Read the direction of transfer

In $P(E' \rightarrow E)$, which energy is before the collision? Why does $E' < E$ describe an energy gain?

5 A first approximation to the fast spectrum

The **uncollided flux** in the fast range provides a first approximation to the fast spectrum. We label this approximation $\phi_{\text{fast}}(E)$ here.

Uncollided fast-spectrum approximation

$$\phi_{\text{fast}}(E) \approx \frac{\chi(E)}{\Sigma_t(E)} s_f''''.$$

FNRP 3.16

The fission-source strength is

$$s_f'''' = \int_0^\infty v(E') \Sigma_f(E') \phi(E') dE'.$$

The flux depends on the flux!

Where does ϕ appear on the right-hand side? Which factors set the spectral shape, and which quantity sets its amplitude?

Scattering changes the fast spectrum. Any light nuclei cause energy degradation through elastic scattering. Inelastic scattering with heavy nuclei is also very important.

Board sketch: inelastic-scattering energy losses

Sketch the inelastic-scattering energy changes, ΔE .

6 Epithermal slowing down: assumptions

Return to the spectrum balance and assume:

1. $E < 0.1$ MeV: no direct fission births at E .
2. No absorption.
3. No external source term in this energy range.

Absorption is deferred to the next lesson.

Scattering-only balance

$$\Sigma_s(E) \phi(E) = \int_0^\infty P(E' \rightarrow E) \Sigma_s(E') \phi(E') dE'.$$

FNRP 3.21

OpenMC example / observations

7 The energy-transfer kernel and its limits

For a single moderator species of mass number A , recall

$$\alpha = \frac{(A-1)^2}{(A+1)^2}.$$

Elastic-scattering energy distribution

$$P(E' \rightarrow E) = \begin{cases} \frac{1}{(1-\alpha)E'}, & \alpha E' < E < E', \\ 0, & \text{otherwise.} \end{cases}$$

Note the switched prime! E' is now the incident energy; E is the outgoing energy at which we write the balance.

Exercise: determine the integration limits

With E fixed, solve $\alpha E' < E < E'$ for the incident energies E' that can contribute. For $\alpha > 0$,

$$\text{_____} < E' < \text{_____}.$$

The resulting reduced balance is

$$\Sigma_s(E) \phi(E) = \int_E^{E/\alpha} \frac{\Sigma_s(E') \phi(E')}{(1-\alpha)E'} dE'.$$

8 Spectrum and slowing-down density

Forms to examine

Candidate shape to verify:

$$\phi(E) = \frac{C}{E}.$$

In terms of the slowing-down density:

$$\phi(E) = \frac{q}{\xi \Sigma_s(E) E}.$$

Here q is the **slowing-down density** and ξ is the slowing-down decrement. In the absence of absorption,

$$q = s_f''''.$$

Verification and consistency check

Substitute the proposed C/E form into the reduced balance. Compare it with $q/[\xi \Sigma_s(E) E]$: what must be true of $\Sigma_s(E)$ for both forms to describe the same flux with constant C , q , and ξ ?